

Data Scientist with expertise in data analysis, machine learning, and full stack application development, passionate about delivering creative, data-driven solutions to complex problems and generating actionable insights.

Educational Background

Princeton University, Princeton, New Jersey	Sep. 2016 – May 2021
<i>Electrical and Computer Engineering</i>	
Ph.D. in Electrical and Computer Engineering	Dissertation Advisor: Prof. H. Vincent Poor
M.A. in Electrical Engineering	GPA: 3.98/4.00
Orta Doğu Technical University, Ankara, Turkey (Double-Major)	Sep. 2009 – May 2014
<i>Mathematics,</i>	Bachelor of Science, GPA: 3.96/4.00
<i>Electrical and Electronics Engineering,</i>	Bachelor of Science, GPA: 3.96/4.00

Patents

Machine Learning-Based Item Reranking Based on User Query and Cart Context	Patent No.: US20250342516A1	2025
Method and System for Fulfillment-Aware Product Search Ranking	(under review)	2026
Automatic Configuration Management in Cellular Networks	(under review)	2022

Coding and Technical

Programming Languages:	Python (XGBoost, Keras, PyTorch, sklearn), Java, JavaScript, React, React Native, bash.
Data & Analytics:	SQL, NoSQL, BigQuery, R, Python (Pandas, PySpark, Matplotlib), Tableau.
Machine Learning & AI:	Neural Networks, Large Language Models, Natural Language Processing (NLP), Classification, Bayesian Inference, Clustering, Regression Analysis.
Development & Platforms:	Google Cloud (GCP), BigQuery, Firestore, Azure, Amazon Web Services, Axios HTTP.
Quantitative Skills:	Probability Theory, Optimization, Data Visualization.

Work Experience

Walmart Global Tech, Staff Data Scientist	Oct. 2021 – Present
Lead end-to-end machine learning (ML) model development for e-commerce Search and BuyBox ranking systems, integrating engineering and data science capabilities across Python and Java. Conduct advanced experimentation, signal design, model training, and runtime deployment to drive large-scale business outcomes. Inform strategy through rigorous A/B testing, support infrastructure improvements, and mentor junior data scientists on navigating Walmart’s complex systems.	
Senior Data Analytics & Machine Learning Engineer at Search Ranking:	
- Delivered a 2.4% GMV lift by architecting a unified ranking system that harmonized default and personalized search rankers, replacing fragmented legacy models with a high-performance, scalable framework. - Improved query-item relevance by 11% by engineering a novel token-importance signal; spearheaded the end-to-end integration via Java-based CI/CD pipelines and advanced ML model deployments. - Directed cross-functional model validation using interleaving and A/B testing frameworks, ensuring statistical rigor and driving consensus among senior stakeholders for high-stakes production changes. - Optimized system reliability and latency by deploying high-throughput ranking features within core infrastructure.	
Staff Data Scientist at Buy Box Offer Ranking:	
- Spearheaded Product Category feature development, driving a site-wide 0.87% lift in Multi-Offer Add-To-Cart (ATC) rates, generating an estimated \$58M in incremental annual revenue. - Increased ATC conversions by 4% by architecting a custom Constrained Neural Network to enforce price and shipping monotonicity, ensuring high-precision ranking logic. - Enhanced multi-offer conversion rates by 3.8% by modeling multi-dimensional relationships between shipping speed, pricing, and buyer intent to optimize item-page rankings. - Automated end-to-end model training and deployment pipelines, significantly reducing iteration time and enabling seamless retraining and rapid experimentation cycles.	
Technologies: Java (Runtime & Unit Testing), Python (PyTorch, Keras, Pandas, XGBoost), Google Cloud (GCP), BigQuery, Azure, Tableau, SQL.	

Personal Projects and Experiences

Portfolio Optimization

May 2025 – Present

- Designed and implemented a portfolio optimization algorithm leveraging an Adaptive Kalman Filter to dynamically adjust asset weights.
- Achieved 4–6% improvement in information coefficient and an 8% reduction in portfolio volatility during backtesting.
- Evaluated performance across multiple market regimes to assess robustness and adaptability.

Technologies: Python (numpy, pandas, yfinance, financetoolkit, financedatabase), Interactive Brokers Paper Trading.

Etiqueta – AI Data Labeling Platform, Founder & Lead Developer

Mar. 2023 – May 2025

Architected and deployed **Etiqueta**, a first-of-its-kind mobile-native platform for AI-aided data labeling and image segmentation. Engineered the full technology stack to enable scalable, gamified crowdsourcing across iOS, Android, and ARM-based desktop environments.

Platform Impact & Open Source Contribution

- **Curated 100K+ high-fidelity labels** and **13K image segmentations** via gamified workflows; published open-source datasets on Kaggle and Hugging Face (1K+ downloads).
- **Published a technical report** and codebase on GitHub, establishing a framework for mobile-first dataset distribution and research adoption.
- **Launched on Global App Stores** as the only public, mobile-native data labeling solution, managing the full product lifecycle from UX design to store optimization.

Full-Stack Engineering

- **Frontend:** Developed a high-throughput React Native interface supporting 30+ concurrent label interactions per screen, optimizing UI responsiveness and reducing interaction latency.
- **Backend:** Built a scalable Python (FastAPI) and Firestore architecture to handle real-time task distribution, utilizing OAuth for secure authentication and end-to-end data privacy.
- **Reliability:** Implemented comprehensive backend unit testing (pytest) and CI/CD practices to ensure cross-platform stability across iOS, Android, and macOS.

Technologies: Python (FastAPI), React Native, JavaScript, Firestore (NoSQL), OAuth, Axios, Pytest.

Awards and Honors

Excellence in Teaching Award at Princeton University

2018

Turkish Presidential Fellowship Award

2009 – 2014

Ranked 78th out of 1.5M+ candidates in the Turkish National University Entrance Exam

2008

Ph.D. Research Highlights

Bounded Gaussian Mean Estimation, 50-year old open problem

- Used non-linear optimization to prove that the least favorable distribution has finite support.
- My technique based on oscillation theorem has been extensively used in various multi access and wiretap channel setups.

Information Secrecy and User Privacy

- Developed a statistical method, called poissonization, to simplify the analysis of weakly dependent random variables.
- My ‘poissonization’ proof technique, has been employed in large deviations and quantum information theory research.

Privacy in Machine Learning

- Used supervised and adversarial machine learning tools to remove sensitive information from data.
- Ensured minimal computational burden on the user to make the solution feasible for all types of ML prediction tasks.

Extra Curricular

Languages: Fluent in English, French, Spanish, and Turkish (Native).

Mentorship: Volunteer mentor at Princeton GradFutures program; each year mentoring about 4 students.

Leadership: Regional Princeton Graduate Alumni Leader; organizing networking events for alumni.

Professional Services

Technical Program Committees:

- IEEE 54th Annual Conference on Information Sciences and Systems (CISS 2020), Princeton, NJ, USA, March 18-20, 2020.
- IEEE 52th Annual Conference on Information Sciences and Systems (CISS 2018), Princeton, NJ, USA, March 22-23, 2018.

Publications

Ongoing Work:

- S. Yagli, “Etiqueta: AI-Aided, Gamified Data Labeling to Label and Segment Data,” *Manuscript prepared*.
- A. Dytso, S. Yagli, and L. Barletta (Eds.), “Collected Works of Hans Sylvain Withenhausen,” *Manuscript prepared*.

Journal Papers:

- M. Al, S. Yagli, and S. Y. Kung, “Privacy Enhancing Machine Learning via Removal of Unwanted Dependencies,” *IEEE Transactions on Neural Networks and Learning Systems*, Sep. 2021.
- A. Dytso, S. Yagli, H. V. Poor, and S. Shamai (Shitz), “Capacity Achieving Distribution for the Amplitude Constrained Additive Gaussian Channel: An Upper Bound on the Number of Mass Points,” *IEEE Transactions on Information Theory*, vol. 66, no. 4, pp. 2006-2022, Apr. 2020.
- S. Yagli and P. Cuff, “Exact Exponent for Soft Covering,” *IEEE Transactions on Information Theory*, vol. 65, no. 10, pp. 6234-6262, Oct. 2019.
- S. Yagli, Y. Altuğ, and S. Verdú, “Minimax Rényi Redundancy,” *IEEE Transactions on Information Theory*, vol. 64, no. 5, pp. 3715-3733, May 2018.

Conference Proceedings:

- S. Yagli, A. Dytso, and H. V. Poor, “Information-Theoretic Bounds on the Generalization Error and Privacy Leakage in Federated Learning,” in *Proc. 21st IEEE International Workshop On Signal Processing Advances in Wireless Communications (SPAWC 2020)*, Atlanta, GA, USA, May 2020.
- S. Yagli, A. Dytso, and H. V. Poor, “Estimation of Bounded Normal Mean: An Alternative Proof for the Discreteness of the Least Favorable Prior,” in *Proc. 2019 IEEE Information Theory Workshop (ITW)*, Visby, Sweden, Aug. 2019.
- S. Yagli, A. Dytso, H. V. Poor, and S. Shamai (Shitz), “An Upper Bound on the Number of Mass Points in the Capacity Achieving Distribution for the Amplitude Constrained Additive Gaussian Channel,” in *Proc. 2019 IEEE International Symposium on Information Theory (ISIT)*, Paris, France, Jul. 2019, pp. 1907-1911.
- S. Yagli, A. Dytso, H. V. Poor, and S. Shamai (Shitz), “Some Aspects of Totally Positive Kernels Useful in Information Theory,” in *Proc. 2019 MoTION Workshop of Wireless Communications and Networking Conference (WCNC)*, Marrakech, Morocco, Apr. 2019, pp. 1-6.
- S. Yagli and P. Cuff, “Exact Soft-Covering Exponent,” in *Proc. 2018 IEEE International Symposium on Information Theory (ISIT)*, Vail, CO, Jun. 2018, pp. 1680-1684.
- S. Yagli, Y. Altuğ, and S. Verdú, “Minimax Rényi Redundancy,” in *Proc. 2017 IEEE International Symposium on Information Theory (ISIT)*, Aachen, Germany, Jun. 2017, pp. 2980-2984.

Dissertation:

- S. Yagli, “Topics in Information and Estimation Theory: Parameter Estimation, Lossless Compression, Constrained Channels, and Error Exponents,” PhD Dissertation, Princeton University, 2021.

Technical Reports:

- S. Temel, S. Yagli, and S. Gören, “P, PD, PI, PID Controllers,” Middle East Technical University, Ankara, Turkey, Technical Report, 2013.